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Instrument 1

Long 500 units – American Option on S&P 500

Strike: 6935 | Expiry: Jun 30, 2026 | Discounting: USD SOFR | Reporting Currency: EUR

I. Risk Factor Identification – VaR vs. RNIV

A. Risk Factors to be Included in VaR

This position represents a nonlinear equity derivative with embedded FX translation risk and interest rate discounting exposure. Therefore, the VaR framework must capture all first- and second-order sensitivities across asset classes.

1. Equity Spot Risk (Delta Exposure)

The primary driver is the S&P 500 spot level. Daily returns of the index must be simulated using either historical or Monte Carlo methods. Given the long position, downside tail risk is the key VaR driver.

2. Equity Volatility Surface (Vega & Skew Risk)

Because the instrument is an option, implied volatility dynamics are critical. The VaR engine must incorporate:

- ATM volatility shifts
- Skew movements (smile dynamics)
- Term structure shifts

Parallel vol shocks are insufficient; surface-level shocks should be applied.

3. Gamma (Convexity Risk)

Near-the-money exposure around strike 6935 creates non-linear P&L response. A delta-normal approximation would understate risk; therefore, gamma must be captured through full revaluation.

4. Interest Rate Risk (USD SOFR Curve)

The option is discounted using USD SOFR. Curve shifts (parallel and non-parallel) affect present value. While rho exposure may be smaller than delta/vega, it is still modellable and must be included.

5. FX Risk (EUR/USD)

Since the payoff is USD-based but denominated in EUR, daily EUR/USD movements create translation P&L. Correlation between equity and FX during risk-off regimes (USD strengthening when equities fall) amplifies tail losses.

6. Cross-Asset Correlation Effects

Equity–vol negative correlation (leverage effect), equity–FX correlation, and equity–rates interaction must be preserved in simulation to avoid underestimation of joint tail risk.

All of the above have observable market data and liquid hedging markets; hence they belong in VaR.

B. Risks Potentially Classified as RNiV

Certain risks may not be fully captured within standard VaR horizons or data windows:

- **Jump Risk / Gap Risk:** Sudden market crashes beyond historical distribution.
- **Volatility Liquidity Risk:** Surface dislocations during stress.
- **Dividend Forecast Risk:** Changes in forward dividend assumptions.
- **Early Exercise Model Risk:** Numerical approximation errors (e.g., binomial vs. finite difference).
- **Extreme Correlation Breakdown:** Structural shifts not represented in historical sample.

These risks should be documented under model risk governance and addressed via stress testing or capital overlays.

II. P&L Simulation Methodology

Given the American feature and nonlinear payoff, a **full revaluation framework** is required.

Recommended VaR Engine:

- Historical Simulation with full repricing as primary
- Monte Carlo (stochastic volatility) for benchmarking and stress validation

Rationale:

1. **Delta-normal is inappropriate** due to gamma and vega exposure.
2. American optionality requires valuation consistency across scenarios.
3. FX and rates must be jointly simulated to preserve cross-asset correlation.
4. Full repricing captures second-order and cross-gamma effects (Equity × FX).

Each scenario should shock:

- S&P spot
- Vol surface nodes
- USD SOFR curve
- EUR/USD FX

The instrument is then repriced under shocked market states to generate a P&L distribution. Expected Shortfall should complement VaR to capture tail severity.

Instrument 2

European Option on CDX.NA.IG Series 40

Notional: \$100mm | Strike: 100 bps | Physical Settlement into 5Y CDS

I. Risk Factor Identification – VaR vs. RNiV

This is a credit derivative option referencing an index of investment-grade corporates. Spread dynamics and credit correlation are central.

A. Risk Factors to be Included in VaR

1. Index Spread Level (Primary Driver)

Daily movements in CDX.NA.IG Series 40 spread levels must be simulated. Spread widening increases option value (buyer of protection).

2. Spread Term Structure Risk

Although exercised into 5Y CDS, term structure shifts affect valuation and forward spread expectations.

3. Spread Volatility (Implied Credit Vol)

The option value depends on spread volatility; thus, vol-surface shocks must be included.

4. Interest Rate Curve (USD SOFR)

Discounting and CDS PV01 exposure create rate sensitivity.

5. Index Basis Risk

On-the-run vs off-the-run basis between Series 40 and newer series must be considered if hedging uses different series.

6. Correlation Among Constituents

Index spread movements depend on constituent correlations; systemic widening must be reflected in historical spread data.

All of these risk factors are modelable and supported by market data.

B. Risks Potentially Classified as RNiV

Credit markets exhibit tail characteristics that may not be fully captured:

- **Jump-to-Default Clustering**
- **Recovery Rate Shocks**
- **Index Roll Risk (Mar/Sep)**
- **Liquidity Dislocation in CDS Markets**
- **Correlation Breakdown During Crisis**
- **Counterparty Credit Exposure (CVA)**

These risks often materialize under systemic stress and may require stress scenarios beyond historical VaR.

II. P&L Simulation Methodology

Because this is a nonlinear spread option with physical settlement, full repricing is again required.

Recommended Approach:

Historical Simulation on Spread Changes with Full Revaluation

Each historical spread shock scenario should:

- Shift index spread levels
- Adjust implied spread volatility
- Shock USD SOFR curve
- Recalculate CDS PV and option payoff

Rationale:

1. Credit spreads display fat tails and regime shifts.
2. Delta-spread approximation underestimates convexity.
3. Physical settlement into a 5Y CDS requires correct PV01 handling.
4. Spread–rate interaction must be preserved.

Monte Carlo with correlated spread and rate dynamics can be used for benchmarking, but historical full repricing is operationally robust.

Expected Shortfall is particularly important for credit instruments due to asymmetric widening risk.

Comparative Risk Perspective

Instrument 1 (Equity Option) is primarily driven by spot, vol, and cross-asset FX dynamics, with nonlinear equity convexity being the core VaR driver.

Instrument 2 (CDX Option) is driven by spread widening risk, correlation breakdown, and systemic credit events, where jump clustering and liquidity risk are more pronounced.

Both require:

- Full revaluation VaR
- Correlated multi-factor simulation
- Expected Shortfall overlay
- Dedicated stress testing framework

Conclusion:

For both instruments, VaR must capture all modelable market risk factors through full repricing. However, due to nonlinear payoffs and systemic tail exposure, a comprehensive framework should integrate:

- Expected Shortfall (97.5% under FRTB alignment)
- Historical and hypothetical stress scenarios
- RNiV documentation with capital overlay where necessary

The objective is not simply to compute VaR, but to ensure the risk engine remains sensitive to convexity, correlation breakdowns, liquidity stress, and regime shifts, thereby maintaining robustness under extreme yet plausible market conditions.